

Some Set Theory

All sets below are subsets of a set X .

- The union of two sets A and $B = A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- The intersection of two sets A and $B = A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- The relative complement of a set B with respect to a set A or simply the difference of A and $B = A \setminus B = A \cap (X \setminus B) = \{x \mid x \in A \text{ and } x \notin B\}$. Here $B \cap (A \setminus B) = \emptyset$
- The complement of a set $A = X \setminus A = \{x \mid x \notin A\}$. Here $A \cup (X \setminus A) = X$ and $A \cap (X \setminus A) = \emptyset$.
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- The following are equivalent :
 - (a) $A \subseteq B$
 - (b) $A \cup B = B$
 - (c) $A \cap B = A$
 - (d) $X \setminus B \subseteq X \setminus A$
 - (e) $A \cap (X \setminus B) = \emptyset$
 - (f) $B \cup (X \setminus A) = X$
 - (g) $A \cup B = A \cup (B \setminus A)$
- De Morgan's laws
 - $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$
 - $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$
 - If $\{A_i\}_{i \in I}$ be a collection of subsets of X then $X \setminus \left(\bigcup_{i \in I} A_i\right) = \bigcap_{i \in I} (X \setminus A_i)$
 - If $\{A_i\}_{i \in I}$ be a collection of subsets of X then $X \setminus \left(\bigcap_{i \in I} A_i\right) = \bigcup_{i \in I} (X \setminus A_i)$

- $A \subseteq A \cup B$ and $B \subseteq A \cup B$
- $A \cap B \subseteq A$ and $A \cap B \subseteq B$
- $A \times B = \{(x, y) \mid x \in A, y \in B\}$

Let X and Y be any sets and $f : X \rightarrow Y$ be a function.

- $A \subseteq B$ then $f(A) \subseteq f(B)$
- $f(A \cup B) = f(A) \cup f(B)$
- $f(A \cap B) \subseteq f(A) \cap f(B)$
- f is 1-1 iff $f(A \cap B) = f(A) \cap f(B)$
- f is onto iff $Y \setminus f(A) \subseteq f(X \setminus A)$
- f is a bijection iff $f(X \setminus A) = Y \setminus f(A)$
- $f(A) \setminus f(B) \subseteq f(A \setminus B)$
- f is 1-1 iff $f(A) \setminus f(B) = f(A \setminus B)$
- $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$
- $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$
- $f^{-1}(Y \setminus C) = f^{-1}(Y) \setminus f^{-1}(C)$
- $A \subseteq f^{-1}(f(A))$
- $f(f^{-1}(C)) \subseteq C$

- $C \subseteq D$ implies $f^{-1}(C) \subseteq f^{-1}(D)$
- If f is 1-1 then $A = f^{-1}(f(A))$
- If f is onto then $C = f(f^{-1}(C))$
- If f is 1-1 then $f(A) = f(B)$ implies that $A = B$
- If f is onto then $f^{-1}(C) = f^{-1}(D)$ implies that $C = D$
- If $\{A_i\}_{i \in I}$ is a collection of subsets of X then $f^{-1}(\bigcup_{i \in I} A_i) = \bigcup_{i \in I} f^{-1}(A_i)$ and

$$f^{-1}(\bigcap_{i \in I} A_i) = \bigcap_{i \in I} f^{-1}(A_i)$$